Software Requirements Specification: KitchenFlow Restaurant Order & Kitchen Display System

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1. Introduction

1.1 Purpose

This document specifies what KitchenFlow shall do and how well it shall do it. It is document 2 of 7 and the contract between the product idea (document 1) and the design documents (3 to 6): every design element and every test must trace back to a requirement ID in this document.

1.2 Scope

KitchenFlow is a web-based order and kitchen-display system for one quick service restaurant (QSR) outlet. It covers counter order entry with pricing and GST, simulated payment recording, automatic routing of items to kitchen stations, a real-time kitchen display with start, bump and recall, a customer status board, lightweight inventory, an audit trail and management reports.

Out of scope for version 1: real payment gateways, delivery-aggregator integrations, multi-outlet or multi-tenant operation, printer or POS hardware integration, customer accounts and personal data, demand forecasting and machine learning, and native mobile apps.

1.3 Intended readers

The developer (implementation and testing), reviewers and interviewers (to judge engineering depth), and anyone writing the later documents.

1.4 Conventions

- **IDs:** FR-MODULE-nn functional requirement; BR-nn business rule; NFR-CATEGORY-nn quality requirement; AT-nn acceptance scenario; OI-nn open issue.

- **Priority:** M (Must, written as shall), S (Should, written as should), C (Could, written as may). Priorities follow the MoSCoW scope in document 1.
- **Verification:** T automated test, D demonstration, I inspection of code, configuration or documents.
- Every requirement is atomic and testable. Money is in INR with two decimals. Times are shown in Asia/Kolkata.

1.5 References

1. KF-PDD-001, Product Development Document, v0.1.
2. ISO/IEC/IEEE 29148, Requirements engineering (used as a structural guide).
3. Glossary in section 15.

2. Overall description

2.1 Product perspective

KitchenFlow is a new, self-contained system built as a modular monolith. Clients are web screens (cashier POS, kitchen station boards, manager board, customer status board, admin screens) that talk to one REST and server-sent-events API backed by a relational database. Version 1 has no integration with external systems.

2.2 Product functions

Module	Code	Summary
Authentication and users	AUTH	Login, roles, station binding
Menu	MENU	Categories, items, modifiers, availability, combos
Orders	ORD	Order entry, pricing, lifecycle, cancellation, search
Payments	PAY	Simulated payment recording, bill, refund entry
Kitchen display	KDS	Tickets per station, start, bump, recall, timers, push
Status board	BRD	Public token board
Inventory-lite	INV	Stock reservation and automatic unavailability
Reporting	RPT	Sales and kitchen-timing reports in SQL

Module	Code	Summary
Audit	AUD	Append-only event log
Configuration	CFG	Stations, tax, thresholds, seed data
Platform	SYS	API conventions, documentation, health, logging

2.3 Operating environment

- Server: Java 17 or newer (Java 21 preferred) on Windows, Linux or macOS, or in a Docker container.
- Database: PostgreSQL 15 or newer (MySQL 8 is an acceptable alternative; the choice is closed in document 3).
- Clients: current Chrome, Edge or Firefox on a desktop or a 10-inch landscape tablet, on the same local network as the server.

2.4 Design and implementation constraints

- Java, Spring Boot and SQL are fixed by the project's purpose.
- All money arithmetic uses decimal types, never floating point.
- Every state-changing operation is transactional and audited.
- The system stores no customer personal data.
- One developer and a fixed time budget, so the Must list is protected first.

2.5 Design-basis workload

One outlet, 800 to 1,500 orders per day, a peak of about 150 orders per hour, up to 5 station screens, 10 staff users at once and 1 status board. The load tests in section 8 deliberately exceed this peak.

3. Users and permissions

3.1 User classes

Class	Description
Cashier	Takes orders and payments at the counter; hands over finished orders
Kitchen	Works at one or more stations; starts, bumps and recalls tickets
Manager	Runs the shift: sees all stations, cancels in-kitchen orders, sets rush, toggles availability, adjusts stock, reads reports and audit
Admin	Configures menu, stations, tax, thresholds and users
Public (customer)	Reads the status board only; never logs in

3.2 Permission matrix

Capability	Cashier	Kitchen	Manager	Admin	Public
Log in	Yes	Yes	Yes	Yes	No
View menu	Yes	Yes	Yes	Yes	No
Manage categories, items, modifiers, stations, tax, configuration	No	No	No	Yes	No
Mark item available or unavailable	No	No	Yes	Yes	No
Create or edit an order (while CREATED)	Yes	No	Yes	No	No
Record payment	Yes	No	Yes	No	No
Cancel an order in CREATED	Yes	No	Yes	No	No
Cancel an order in IN_KITCHEN	No	No	Yes	No	No
View station tickets	No	Own stations	All	No	No
Start, bump or recall tickets	No	Own stations	All	No	No
Mark order handed over	Yes	No	Yes	No	No

Capability	Cashier	Kitchen	Manager	Admin	Public
Flag order as rush	No	No	Yes	No	No
Adjust stock	No	No	Yes	Yes	No
View reports	No	No	Yes	Yes	No
View audit log	No	No	Yes	Yes	No
Manage users and roles	No	No	No	Yes	No
View status board	No	No	No	No	Yes (read only)

The status board is open to anyone on the network and exposes only token numbers and statuses.

4. Business rules

ID	Rule
BR-01	Token numbers are unique within a business day, start at 1, increase by one per order and restart at the business-day cutoff (BR-16).
BR-02	All money uses decimal arithmetic in INR with two decimals. Each order line stores a snapshot of the unit price and GST rate used, so later menu changes never alter existing orders.
BR-03	GST per line = (unit price plus selected modifier prices) x quantity x category rate, rounded half-up to two decimals. Order total = sum of line amounts plus line GST. Menu prices are exclusive of GST (see OI-05).
BR-04	An unavailable or inactive item cannot be added to an order. Orders already created are unaffected.
BR-05	A CREATED order may be cancelled by a Cashier or Manager. An IN_KITCHEN order may be cancelled by a Manager only. READY and COMPLETED orders cannot be cancelled. A reason is always required.

ID	Rule
BR-06	An order is READY exactly when all of its non-cancelled tickets are DONE. Recalling a ticket of a READY order returns the order to IN_KITCHEN.
BR-07	Payment is taken before the kitchen sees the order (pay first). Partial and split payments are not supported in version 1.
BR-08	The payment amount must equal the order total.
BR-09	Every item maps to at least one station. An item mapped to several stations appears on the ticket of each of those stations. A combo has no station of its own; each component routes to its own stations.
BR-10	Station queues are ordered by RUSH first, then by payment time, oldest first.
BR-11	Ticket colour: green until 80 percent of the target time has elapsed, amber from 80 to 100 percent, red beyond 100 percent. A ticket's target is the longest target time among its items. Thresholds are configurable.
BR-12	Stock is reserved when an order is created and restored when the order is cancelled while CREATED. Cancelling an IN_KITCHEN order does not restore stock (treated as wastage). An item with tracked stock becomes unavailable at zero.
BR-13	Every state change, payment, price change, availability change and user or role change creates an audit record in the same transaction.
BR-14	An order can be edited only while CREATED.
BR-15	A request that repeats an Idempotency-Key within 24 hours with an identical payload returns the original response. With a different payload it is rejected with 422 IDEMPOTENCY_KEY_REUSED.

ID	Rule
BR-16	The business day runs from the configured cutoff (default 04:00, Asia/Kolkata) to the next cutoff. Tokens and reports use the business day.
BR-17	A DONE ticket can be recalled only within the configured recall window (default 10 minutes) and only while its order is not COMPLETED.
BR-18	Timestamps are stored in UTC and displayed in Asia/Kolkata.

5. Functional requirements

Columns: ID, requirement, priority (Pri), acceptance criteria. Numbers in acceptance criteria are targets to verify, not guesses.

5.1 Authentication and users (AUTH)

ID	Requirement	Pri	Acceptance criteria
FR-AUTH-01	The system shall authenticate users by username and password and issue a signed access token valid for 15 minutes.	M	Valid credentials return 200 and a token. A wrong username or wrong password returns 401 with the same message.
FR-AUTH-02	The system shall store passwords only as BCrypt hashes and shall never log or return them.	M	Database inspection shows no plain text; searching the logs for a known test password finds nothing.
FR-AUTH-03	The system shall enforce role-based authorization on every endpoint except login, health and the public status board.	M	One test per endpoint and role matches section 3.2. Unauthenticated calls return 401; unauthorized calls return 403.
FR-AUTH-04	The system shall bind each Kitchen user to one or more stations and let that user view and act only on those stations' tickets.	M	A Fry-station user requesting Grill tickets receives 403.

ID	Requirement	Pri	Acceptance criteria
FR-AUTH-05	The Admin shall create, deactivate and reactivate users and assign one role per user.	M	A deactivated user cannot log in; tokens already issued expire within 15 minutes.
FR-AUTH-06	The system should lock an account for 15 minutes after 5 consecutive failed logins within 10 minutes.	S	The sixth attempt returns 429, even with the correct password, during the lockout.
FR-AUTH-07	The system should support refresh tokens that rotate on every use.	S	Reusing a consumed refresh token returns 401.

5.2 Menu (MENU)

ID	Requirement	Pri	Acceptance criteria
FR-MENU-01	The Admin shall create, edit and deactivate categories with name, display order and GST rate.	M	Rate must be between 0 and 28 percent; a duplicate name returns 422.
FR-MENU-02	The Admin shall create, edit and deactivate items with name, description, category, base price, one or more stations, target preparation time in seconds and optional stock tracking.	M	An item with no station is rejected (BR-09); price must be greater than 0.
FR-MENU-03	The Admin shall define modifier groups per item with selection type (single or multiple), minimum and maximum selections, and options with an optional price change.	M	An order that violates a minimum or maximum is rejected with field-level errors (FR-ORD-02).

ID	Requirement	Pri	Acceptance criteria
FR-MENU-04	The Manager and Admin shall mark an item unavailable or available, effective for new orders immediately.	M	After marking unavailable, creating an order with that item returns 422 ITEM_UNAVAILABLE within 1 second.
FR-MENU-05	The system shall never hard-delete an item that appears in any order; it shall deactivate it.	M	Deleting a used item returns 409; the item leaves the POS menu but stays in order history.
FR-MENU-06	The system shall provide a POS menu endpoint that returns active items grouped by category with availability and modifier groups.	M	Inactive items are excluded; filter by category and text search work; p95 under 300 ms.
FR-MENU-07	The system should support combos: a named bundle of items at a fixed price that expands into component items for routing.	S	A combo order creates tickets for each component's station and the bill shows one combo line.
FR-MENU-08	The system shall apply a price change only to orders created after it.	M	An order created before a price change keeps its original line prices.

5.3 Orders (ORD)

ID	Requirement	Pri	Acceptance criteria
FR-ORD-01	The Cashier and Manager shall create an order with an order type (DINE_IN, TAKEAWAY, DELIVERY), one or more lines (item, quantity 1 to 50, selected modifiers) and an optional note of up to 200 characters.	M	A valid request returns 201 with order number, token and totals.
FR-ORD-02	The system shall validate every order and return all validation errors together, each with its field path.	M	A request with an inactive item and an invalid quantity returns one 422 listing both.

ID	Requirement	Pri	Acceptance criteria
FR-ORD-03	The system shall calculate line totals, GST per line and the order total using BR-02 and BR-03 and return the breakdown.	M	Test vectors with modifiers and mixed GST rates match hand-calculated values to the paisa.
FR-ORD-04	The system shall generate a token (BR-01) and a unique, human-readable order number.	M	1,000 concurrent creations yield unique tokens within the business day.
FR-ORD-05	The system shall enforce the order lifecycle in section 6.1 and reject every other transition with 409 INVALID_TRANSITION.	M	One test per illegal transition passes.
FR-ORD-06	The Cashier and Manager shall edit an order's lines or note while it is CREATED; edits shall recalculate totals and adjust stock.	M	After payment, an edit returns 409.
FR-ORD-07	The system shall cancel an order under BR-05 with a mandatory reason code, cancel its open tickets and handle stock under BR-12.	M	A cancel without a reason returns 422; open tickets become CANCELLED.
FR-ORD-08	The Cashier and Manager shall mark a READY order COMPLETED when it is handed over.	M	Marking an IN_KITCHEN order COMPLETED returns 409.
FR-ORD-09	The system shall search orders by status, date range, token, order number and type with pagination (default 20, maximum 100) and newest-first sorting.	M	Result counts match the filters on a 10,000-order dataset; p95 under 300 ms.
FR-ORD-10	The system should make order creation idempotent through an Idempotency-Key header (BR-15).	S	50 concurrent identical requests produce one order and identical responses.
FR-ORD-11	The Manager shall flag or unflag an open order as RUSH.	S	A flagged order sorts first on every station queue.

ID	Requirement	Pri	Acceptance criteria
FR-ORD-12	The system should cancel orders left unpaid in CREATED longer than a configured timeout (default 15 minutes) and restore their stock.	S	After the timeout the order is CANCELLED with reason UNPAID_TIMEOUT and an audit record.

5.4 Payments (PAY)

ID	Requirement	Pri	Acceptance criteria
FR-PAY-01	The Cashier and Manager shall record a payment for a CREATED order with method (CASH, UPI, CARD), amount and an optional reference of up to 64 characters.	M	Success returns 201 and the order moves to PAID.
FR-PAY-02	On successful payment the system shall create the kitchen tickets and move the order to IN_KITCHEN in the same database transaction.	M	A fault injected during ticket creation leaves the order CREATED and no payment stored.
FR-PAY-03	The system shall allow exactly one successful payment per order.	M	Two concurrent payment requests give one 201 and one 409 PAYMENT_ALREADY_RECORDED; a unique constraint exists in the database.
FR-PAY-04	The system shall reject a payment whose amount differs from the order total.	M	422 AMOUNT_MISMATCH and no state change.
FR-PAY-05	The system shall return a bill with items, modifiers, GST breakdown, total, token, order number, time and payment method.	M	Bill totals equal order totals; GST per rate sums correctly.
FR-PAY-06	For cash payments the system should accept the tendered amount and return the change.	S	Tendered below the total returns 422; change equals tendered minus total.

ID	Requirement	Pri	Acceptance criteria
FR-PAY-07	The system should record a refund entry (status REFUND_PENDING) when a paid order is cancelled.	S	Cancelling an IN_KITCHEN order creates one refund entry for the paid amount.
FR-PAY-08	The system may render the bill as a printable PDF.	C	PDF totals match the JSON bill.

5.5 Kitchen display (KDS)

ID	Requirement	Pri	Acceptance criteria
FR-KDS-01	When an order enters IN_KITCHEN the system shall create one ticket per station involved, holding the items, quantities, modifiers, note, token, order type, rush flag and creation time.	M	An order with fries and a burger creates tickets for the Fry and Grill stations, each with only its own items.
FR-KDS-02	A station screen shall list that station's open tickets (NEW and IN_PROGRESS) in the order of BR-10, with an age timer and the colour of BR-11.	M	Tickets reorder when rush is set; colour changes at 80 and 100 percent of the target time.
FR-KDS-03	A Kitchen user shall START a NEW ticket and BUMP an IN_PROGRESS ticket at a station they are bound to; a Manager may do so at any station.	M	Bumping a NEW ticket returns 409 INVALID_TRANSITION (see OI-01).
FR-KDS-04	A Kitchen user or Manager shall RECALL a DONE ticket within the recall window (BR-17); the ticket returns to IN_PROGRESS.	M	A recall after the window, or after the order is COMPLETED, returns 409.

ID	Requirement	Pri	Acceptance criteria
FR-KDS-05	The system shall push ticket creation, change and cancellation to subscribed station screens in real time using server-sent events, and a reconnecting screen shall reload the full current state.	M	A new ticket appears within 1 s at p95; after a forced disconnect the screen shows all current tickets within 3 s.
FR-KDS-06	The system shall set an order to READY exactly once, in the same transaction as the bump of its last open ticket (BR-06).	M	Two concurrent bumps of an order's last two tickets produce one READY transition and one audit record.
FR-KDS-07	The system shall detect two users changing the same ticket concurrently, accept one change and reject the other with 409 STALE_VERSION and the current ticket state.	M	A two-thread test shows exactly one success.
FR-KDS-08	When an order is cancelled the system shall remove its open tickets from station queues and show a CANCELLED notice until a user acknowledges it.	M	The notice appears on every affected station and clears on acknowledgement.
FR-KDS-09	The manager board shall show every open order with the status of each of its tickets and list tickets that are past their target time.	M	A ticket past target appears in the late list within 5 s.
FR-KDS-10	The system should show a distinct rush marker on rush tickets.	S	Rush tickets are visually distinct and sort first.

5.6 Status board (BRD)

ID	Requirement	Pri	Acceptance criteria
FR-BRD-01	The system shall provide a public, read-only status board that lists token numbers in two groups: PREPARING (order IN_KITCHEN) and READY.	M	The board needs no login and shows no names, items or amounts.
FR-BRD-02	The board shall update within 2 seconds of a status change, using server-sent events with polling as a fallback.	M	A bump that makes an order READY moves its token to READY within 2 s.
FR-BRD-03	The board shall remove a token when its order becomes COMPLETED or CANCELLED.	M	Tokens of completed and cancelled orders are not shown.
FR-BRD-04	The board should highlight tokens that have been READY longer than a configured time (default 5 minutes).	S	A token READY for 6 minutes is highlighted.

5.7 Inventory-lite (INV)

ID	Requirement	Pri	Acceptance criteria
FR-INV-01	The Admin should enable stock tracking per item and set a quantity (integer, 0 or more).	S	Items without tracking are never blocked by stock.
FR-INV-02	At order creation the system should reserve stock for all tracked items atomically; if any item lacks stock, nothing is reserved and the request fails with 409 OUT_OF_STOCK listing the items.	S	100 concurrent orders for an item with stock 10 succeed exactly 10 times and stock never goes below zero.

ID	Requirement	Pri	Acceptance criteria
FR-INV-03	The system should mark an item unavailable when its stock reaches zero and available again when stock is added.	S	The POS menu shows the item unavailable within 1 s of the last unit being reserved.
FR-INV-04	The system should restore reserved stock when an order is cancelled while CREATED (BR-12).	S	Stock after the cancel equals stock before the order.
FR-INV-05	The Manager and Admin should adjust stock with a mandatory reason.	S	An adjustment without a reason returns 422; every adjustment is audited.

5.8 Reporting (RPT)

ID	Requirement	Pri	Acceptance criteria
FR-RPT-01	The Manager and Admin shall obtain a sales summary for a business day or date range: order count, gross amount, GST and total, grouped by payment method and order type.	M	Totals equal the sum of paid, non-cancelled orders computed independently in a test.
FR-RPT-02	The system shall report the top N items by quantity and by revenue for a date range.	M	Ranking matches ground truth on seeded data; ties are broken by name.
FR-RPT-03	The system shall report preparation time per station and per item (average, median, 90th percentile) and the share of tickets that exceeded their target.	M	Values match an independent calculation on seeded timestamps.
FR-RPT-04	The system should report orders per hour of day for a date range.	S	The 24 hourly buckets sum to the total order count.
FR-RPT-05	The system should report cancelled orders grouped by reason and by the cancelling role.	S	Counts match the audit records.

ID	Requirement	Pri	Acceptance criteria
FR-RPT-06	All reports shall be computed in the database with SQL aggregation, not in application memory, and accept a date range of at most 92 days.	M	A range over 92 days returns 422; the query plans are reviewed and use indexes.
FR-RPT-07	The system may export any report as CSV.	C	Exported rows equal the on-screen rows.

5.9 Audit (AUD)

ID	Requirement	Pri	Acceptance criteria
FR-AUD-01	The system shall record, for each state change, payment, cancellation, price change, availability change, stock adjustment and user or role change: actor, role, time, entity type and id, action, previous and new state, and the reason where given.	M	A walk through AT-01 produces one audit record per transition with the correct actor.
FR-AUD-02	Audit records shall be append-only: the API offers no update or delete, and the application's database user has no update or delete permission on the audit table.	M	An update or delete attempted through SQL as the application user fails.
FR-AUD-03	The Manager and Admin shall search the audit log by entity, actor, action and date range with pagination.	M	Filters combine correctly; p95 under 300 ms with 500,000 records.
FR-AUD-04	The system shall write the audit record in the same transaction as the change it describes.	M	If the audit write is forced to fail, the business change is rolled back.

5.10 Configuration and seed data (CFG)

ID	Requirement	Pri	Acceptance criteria
FR-CFG-01	The Admin shall configure stations, GST rates, colour thresholds (BR-11), recall window, business-day cutoff and unpaid-order timeout.	M	A changed value takes effect for new operations without a restart.
FR-CFG-02	The system shall load seed data with one command: at least 5 categories, 30 items with modifiers, 4 stations and one user per role.	M	On an empty database the command produces a ready demo in under 60 seconds.
FR-CFG-03	The system should provide a repeatable script that generates 100,000 synthetic historical orders for performance testing.	S	Reports and list endpoints run against the generated dataset.

5.11 Platform (SYS)

ID	Requirement	Pri	Acceptance criteria
FR-SYS-01	The API shall be versioned under /api/v1, use JSON and return every error in one consistent envelope (section 10).	M	Every error response in the test suite matches the schema.
FR-SYS-02	The system shall generate OpenAPI documentation from the code and serve an interactive page.	M	Every endpoint appears with request, response and error examples.
FR-SYS-03	List endpoints shall support pagination, sorting and filtering with the same parameter names everywhere.	M	A contract test checks the shared parameters on all list endpoints.
FR-SYS-04	The system shall assign a correlation ID to every request, return it in a response header and include it in every log line.	M	A request can be traced across the logs by its ID.

ID	Requirement	Pri	Acceptance criteria
FR-SYS-05	The system shall expose a public health endpoint and an Admin-only metrics endpoint.	M	Health reports database connectivity.

6. State models

6.1 Order lifecycle

From	To	Trigger	Who	Guard
(none)	CREATED	Create order	Cashier, Manager	Items valid and available; stock reserved
CREATED	PAID	Record payment	Cashier, Manager	Amount equals total; no earlier payment
PAID	IN_KITCHEN	Automatic	System	Tickets created in the same transaction
IN_KITCHEN	READY	Last ticket becomes DONE	System	All non-cancelled tickets DONE
READY	IN_KITCHEN	A ticket is recalled	Kitchen (own station), Manager	Inside the recall window
READY	COMPLETED	Hand over	Cashier, Manager	None
CREATED	CANCELLED	Cancel	Cashier, Manager	Reason required; stock restored
CREATED	CANCELLED	Unpaid timeout	System	Timeout elapsed (FR-ORD-12)
IN_KITCHEN	CANCELLED	Cancel	Manager	Reason required; open tickets cancelled

Every other combination is illegal and returns 409 INVALID_TRANSITION. COMPLETED and CANCELLED are final states.

6.2 Ticket lifecycle

From	To	Trigger	Who	Guard
(none)	NEW	Order enters IN_KITCHEN	System	None
NEW	IN_PROGRESS	Start	Kitchen (own station), Manager	None
IN_PROGRESS	DONE	Bump	Kitchen (own station), Manager	None
DONE	IN_PROGRESS	Recall	Kitchen (own station), Manager	Inside the recall window; order not COMPLETED
NEW or IN_PROGRESS	CANCELLED	Order cancelled	System	None

Every ticket carries a version number that changes on each update; it is the basis of FR-KDS-07.

7. External interface requirements

7.1 User interfaces

Screen	Users	Key content
Login	All staff	Username, password, clear error messages
POS order entry	Cashier, Manager	Category tabs, item search, modifiers, running total with GST, note, order type
Payment and bill	Cashier, Manager	Method, amount, change, bill view

Screen	Users	Key content
Order list	Cashier, Manager	Search and filters, status, open an order
Station board	Kitchen	Open tickets, timers and colours, start, bump, recall, cancelled notices
Manager board	Manager	All orders with per-station status, late list, rush toggle, cancel
Status board	Public	PREPARING and READY token columns, full-screen mode
Menu admin	Admin	Categories, items, modifiers, stations, availability, stock
Configuration and users	Admin	Thresholds, GST rates, users and roles
Reports	Manager, Admin	Date range controls, tables, simple charts
Audit log	Manager, Admin	Search and filters

7.2 API interfaces (indicative; the final contract is document 5)

Resource	Endpoints	Notes
Auth	POST /api/v1/auth/login, POST /api/v1/auth/refresh	Bearer token afterwards
Menu	GET /api/v1/menu, PATCH /api/v1/items/{id}/availability, admin CRUD under /api/v1/admin	
Orders	POST /api/v1/orders, GET /api/v1/orders, GET and PUT /api/v1/orders/{id}, POST /api/v1/orders/{id}/cancel, POST /api/v1/orders/{id}/complete, PATCH /api/v1/orders/{id}/rush	Idempotency-Key on creation
Payments	POST /api/v1/orders/{id}/payments, GET /api/v1/orders/{id}/bill	

Resource	Endpoints	Notes
Tickets	GET /api/v1/stations/{id}/tickets, POST /api/v1/tickets/{id}/start, bump and recall	Version required on updates
Streams	GET /api/v1/stream/stations/{id}, GET /api/v1/stream/board	text/event-stream
Board	GET /api/v1/board	Snapshot for first load and reconnect
Inventory	PATCH /api/v1/items/{id}/stock	
Reports	GET /api/v1/reports/sales, top-items, prep-times, hourly, cancellations	
Audit	GET /api/v1/audit	
Admin	/api/v1/admin/users, stations, config	
Health	GET /actuator/health	Public

7.3 Software interfaces

- Relational database through JDBC (PostgreSQL 15 or newer by default).
- Docker and Docker Compose for packaging; GitHub Actions for continuous integration.

7.4 Hardware interfaces

No special hardware. Any device with a current browser works; touchscreen tablets are recommended for stations.

7.5 Communication interfaces

- HTTP with JSON bodies; server-sent events as long-lived GET requests with automatic client reconnect.
- HTTPS outside local development.
- Ticket age is computed from server-provided timestamps, so screens show correct timers even when device clocks differ.

8. Non-functional requirements

Reference environment for performance targets: a laptop-class machine with 4 cores and 8 GB RAM, application and database on the same machine.

ID	Requirement	Verification
NFR-PERF-01	Order creation, including pricing and stock reservation, completes within 300 ms at p95 under 50 concurrent users with 100,000 orders in the database.	T: load test
NFR-PERF-02	A new ticket appears on a connected station screen within 1 s at p95 after the payment transaction commits, with 10 screens connected.	T: timestamp comparison
NFR-PERF-03	List and search endpoints respond within 300 ms at p95 with 100,000 orders.	T: load test
NFR-PERF-04	Each report responds within 2 s at p95 over a 31-day range with 100,000 orders.	T: load test and query-plan review
NFR-CONC-01	50 concurrent identical idempotent create requests create exactly one order.	T
NFR-CONC-02	100 concurrent orders for an item with stock 10 succeed exactly 10 times; stock never goes below zero.	T
NFR-CONC-03	Two concurrent updates to one ticket: one succeeds, the other returns 409 STALE_VERSION, and the ticket stays in a valid state.	T
NFR-CONC-04	Concurrent bumps of the last two tickets of one order produce exactly one READY transition.	T
NFR-SEC-01	Tokens are signed with a key supplied through the environment, never committed, and expire after 15 minutes.	I

ID	Requirement	Verification
NFR-SEC-02	All input is validated on the server; every query is parameterised and no SQL is built from user text.	T and I
NFR-SEC-03	CORS is limited to configured origins, standard security headers are set, and HTTPS is required outside local development.	I
NFR-SEC-04	The system collects no customer personal data; staff data is limited to username, role and password hash.	I
NFR-SEC-05	A review against the OWASP Top 10 is completed and its findings are logged.	I
NFR-REL-01	State-changing operations are atomic; a failure leaves no partial state.	T: fault injection
NFR-REL-02	After an application restart no committed data is lost and station screens reconnect and reload within 3 s.	D
NFR-REL-03	Schema changes are applied only through versioned migrations.	I
NFR-USE-01	Station and board text is readable from 1.5 m: item names at least 24 px, status shown by text and icon as well as colour, touch targets at least 44 px.	I
NFR-USE-02	A person new to the POS screen enters a 3-item order and records payment within 60 seconds after a 5-minute walkthrough.	D: test with 3 people
NFR-MNT-01	Code is packaged by feature and controllers never access repositories directly; an automated architecture test enforces this.	T
NFR-MNT-02	Line coverage is at least 70 percent on service and domain layers, and the build fails below it.	T

ID	Requirement	Verification
NFR-MNT-03	Static analysis and formatting checks run in CI and must pass.	I
NFR-OBS-01	Logs are structured and carry the correlation ID and order number on every order-related line.	I
NFR-OBS-02	Metrics include orders created, ticket cycle time, open stream connections and error count.	I
NFR-POR-01	One docker compose command starts the application and database with seed data on Windows, Linux and macOS.	D
NFR-DOC-01	The README documents how to run, test and demo the system, and the API documentation is generated, not hand-written.	I

9. Data requirements

This is the logical view; tables, types and indexes are finalised in document 4.

Entity	Key attributes	Notes
User	id, username, password hash, role, active	Kitchen users linked to stations
Station	id, name, active	Default: Fry, Grill, Assembly, Beverage
Category	id, name, display order, GST rate, active	
MenuItem	id, name, description, category, base price, target prep seconds, active, available, tracked stock, stock quantity	Linked to one or more stations

Entity	Key attributes	Notes
ModifierGroup, Modifier	group: item, name, selection type, min, max; option: name, price change	
Combo, ComboItem (Should)	combo: name, price; component: item, quantity	
Order	id, order number, token, business day, type, status, rush flag, note, subtotal, GST total, total, version, created and updated times, creator	Unique per business day and token
OrderLine	id, order, item or combo, name snapshot, unit price snapshot, GST rate snapshot, quantity, line GST, line total	Snapshots protect history
OrderLineModifier	line, modifier name snapshot, price change snapshot	
Ticket	id, order, station, status, version, created, started and done times, target seconds	
TicketItem	ticket, order line, quantity	
Payment	id, order, method, amount, reference, tendered amount, status, time, recorded by	One per order
Refund (Should)	id, payment, amount, status, reason	
StockAdjustment (Should)	id, item, change, reason, actor, time	
AuditEvent	id, time, actor, role, entity type, entity id, action, previous state, new state, reason, correlation id	Append-only
IdempotencyRecord (Should)	key, request hash, stored response, created time	Expires after 24 hours

Entity	Key attributes	Notes
Config	key, value	Admin-managed settings

ID	Data rule
DR-01	Money columns use fixed-point decimals (for example DECIMAL(12,2)); no floating-point types.
DR-02	Timestamps are stored in UTC.
DR-03	Every relationship has a foreign key. Check constraints enforce quantity 1 to 50, price above 0 and stock of 0 or more.
DR-04	Unique constraints: token per business day, order number, one payment per order, idempotency key.
DR-05	Menu items and users that are referenced are deactivated, never deleted.
DR-06	Names and prices on order lines are snapshots, not references to live menu values.
DR-07	Indexes are planned for orders by status and time, orders by business day, tickets by station and status, and audit by entity and time; each is justified with an EXPLAIN plan in document 4.
DR-08	Volume basis: 1,500 orders per day, about 8 audit records per order; the performance dataset has 100,000 orders.
DR-09	Orders and audit records are kept for at least 12 months in the demo environment; version 1 deletes nothing.

10. Validation and error handling

Every error response uses one envelope: timestamp, status, code, message, path, correlationId and an errors list of field path and message. Rules: no stack traces or internal details in responses; wrong username and wrong password give the same message;

validation returns all errors at once.

Code	HTTP	When
MALFORMED_REQUEST	400	Body cannot be parsed or has the wrong types
UNAUTHENTICATED	401	Missing, invalid or expired token; failed login
FORBIDDEN	403	Role or station not permitted
NOT_FOUND	404	Unknown id
INVALID_TRANSITION	409	Illegal order or ticket transition
STALE_VERSION	409	Concurrent update lost
OUT_OF_STOCK	409	Insufficient stock
PAYMENT_ALREADY_RECORDED	409	Second payment on an order
VALIDATION_FAILED	422	One or more field rules violated
ITEM_UNAVAILABLE	422	Unavailable or inactive item in an order
AMOUNT_MISMATCH	422	Payment amount differs from the order total
IDEMPOTENCY_KEY_REUSED	422	Same key with a different payload
RATE_LIMITED	429	Login lockout or throttling
INTERNAL_ERROR	500	Unexpected failure; generic message, details only in the logs

11. Constraints, assumptions and dependencies

Assumptions

- A-01: one outlet and one time zone (Asia/Kolkata).
- A-02: menu prices are exclusive of GST (open issue OI-05).
- A-03: payments are simulated; no money moves.
- A-04: the server and all screens share a stable local network.
- A-05: four stations by default (Fry, Grill, Assembly, Beverage), configurable by the Admin.

- A-06: staff are trained on the screens; one Manager is on shift at a time.

Dependencies: a JDK, a PostgreSQL instance, Docker, a current browser and a CI service. There are no external APIs.

Change control: any change to a Must requirement is recorded in the revision history with its reason, and the affected tests in document 7 are updated in the same change.

12. Verification and traceability

12.1 Traceability to product goals

Goal (document 1)	Requirements
G1 Correctness under rush	FR-ORD-10, FR-PAY-03, FR-KDS-06, FR-KDS-07, FR-INV-02, NFR-CONC-01 to 04
G2 Real-time visibility	FR-KDS-05, FR-BRD-02, NFR-PERF-02, NFR-REL-02
G3 Lifecycle and audit	FR-ORD-05, FR-KDS-03, FR-KDS-04, FR-AUD-01 to 04, BR-05, BR-06, BR-13, sections 6.1 and 6.2
G4 Measurable kitchen performance	FR-RPT-01 to 06, NFR-PERF-04
G5 Easy to run and demonstrate	FR-CFG-02, FR-CFG-03, NFR-POR-01, NFR-DOC-01
G6 Defensible engineering	NFR-SEC-01 to 05, NFR-MNT-01 to 03, NFR-OBS-01 and 02, FR-SYS-01 to 05

12.2 Acceptance scenarios

ID	Scenario	Expected result	Covers
AT-01	Cashier creates a 3-item order with a modifier and pays by UPI; Fry and Grill tickets appear; cooks start and bump; the order is handed over	Order COMPLETED; token moved from PREPARING to READY and then off the board; bill correct; one audit record per transition	FR-ORD-01 to 05, FR-PAY-01 and 02, FR-KDS-01 to 06, FR-BRD-01 to 03, FR-AUD-01

ID	Scenario	Expected result	Covers
AT-02	Submit an order with an unavailable item and quantity 0	One 422 listing both errors; no order created	FR-ORD-02, FR-MENU-04
AT-03	Send the same create request 50 times concurrently with one key	One order and 50 identical responses	FR-ORD-10
AT-04	100 concurrent orders for an item with stock 10	Exactly 10 succeed; stock is 0; the item shows unavailable	FR-INV-02, FR-INV-03
AT-05	Two users bump the same ticket at once	One succeeds; the other gets 409 STALE_VERSION	FR-KDS-07
AT-06	Bump the last ticket, then recall it inside the window	Order goes READY, then back to IN_KITCHEN; board follows	FR-KDS-04, BR-06
AT-07	Cashier cancels a CREATED order; cashier tries to cancel an IN_KITCHEN order; Manager cancels it; Manager tries a READY order	Allowed; 403; allowed with reason; 409	FR-ORD-07, BR-05
AT-08	Each role calls every endpoint it should not reach	401 or 403 every time	FR-AUTH-03, FR-AUTH-04
AT-09	Cut a station screen's connection for 30 seconds while orders arrive	After reconnect the screen shows all current tickets within 3 s	FR-KDS-05
AT-10	Run every report on the seeded dataset	Figures equal independent SQL ground truth	FR-RPT-01 to 06
AT-11	Force an error during ticket creation after payment	Order stays CREATED; no payment and no partial tickets	FR-PAY-02, NFR-REL-01

Every requirement receives at least one test case ID in document 7.

13. Learning map for the developer

The SRS is written to be complete, not trimmed to current skill. This map lists what you need to learn to implement it, with the requirements each topic unlocks. Core topics are needed for the Must list; Stretch topics add depth for the Should list and interviews.

How to learn each topic: read or watch, build a one-hour throwaway spike, explain the trade-off in two sentences, then apply it in the project and write a test. A topic is learned when you can redo the spike without a tutorial and explain why you chose it. Learn a topic when its phase starts, not all at once.

Topic	Needed for	What to be able to do	Level	Phase
Java essentials	Everything	Classes, interfaces, enums with behaviour, records, generics, collections, streams, Optional, exceptions, immutability	Core	0-1
Money and time in Java	BR-02, BR-03, BR-16, BR-18	BigDecimal with scale and RoundingMode.HALF_UP (never double); java.time with Instant and ZoneId Asia/Kolkata	Core	1
Spring Boot fundamentals	All FR	Dependency injection, configuration properties and profiles, REST controllers, Bean Validation, global error handling with @ControllerAdvice	Core	1
SQL and relational design	Section 9, RPT	Primary and foreign keys, unique and check constraints, 3NF, indexes, joins, GROUP BY, window functions, percentile functions, CTEs, reading EXPLAIN	Core	0-3
Database migrations	NFR-REL-03	Flyway versions and a baseline strategy	Core	0-1

Topic	Needed for	What to be able to do	Level	Phase
JPA and Hibernate	Persistence	Entity mapping, relationships, lazy versus eager loading, the N+1 problem, fetch joins, DTO projections, native queries	Core	1
Transactions and isolation	FR-PAY-02, FR-AUD-04, NFR-REL-01	ACID, @Transactional propagation and rollback rules, isolation levels and what each prevents	Core	1-3
State machine design	Sections 6.1 and 6.2	Transition tables or the State pattern with enums, guard conditions, a test for every legal and illegal transition	Core	1
Spring Security and JWT	AUTH, NFR-SEC	Authentication versus authorization, filter chain, BCrypt, JWT structure and validation, method security, CORS, OWASP Top 10 (broken access control, injection)	Core	1
REST API design and OpenAPI	SYS, section 7	Resource naming, status codes, pagination, one error envelope, springdoc	Core	1
Concurrency control	FR-KDS-06 and 07, FR-PAY-03, FR-INV-02, NFR-CONC	Race conditions, optimistic locking with @Version, pessimistic locks (SELECT FOR UPDATE), atomic conditional UPDATE, unique constraints as guards, multithreaded tests with ExecutorService and CountDownLatch. This is the hardest and most valuable topic	Core	2-3

Topic	Needed for	What to be able to do	Level	Phase
Real-time with server-sent events	FR-KDS-05, BRD	How SSE works, SseEmitter lifecycle (timeouts, errors, cleanup), event ids, reconnect with full-state resync, why SSE rather than WebSocket here	Core	2
Idempotency pattern	FR-ORD-10, BR-15	Idempotency keys, request hashing, storing and replaying responses	Core	3
Testing	All T items, NFR-MNT	JUnit 5, AssertJ, Mockito, Spring Boot test slices, Testcontainers with PostgreSQL, test data builders, basic load testing with k6	Core	1-3
Docker and CI	NFR-POR, NFR-MNT-03	Multi-stage Dockerfile, Compose with environment configuration, a GitHub Actions workflow with build, tests and a coverage gate	Core	0-4
Observability	NFR-OBS, FR-SYS-04	SLF4J and Logback, structured logs, MDC for the correlation id, Actuator and Micrometer metrics	Core	3
Frontend basics	Section 7.1	HTML, CSS, JavaScript fetch and EventSource, rendering state; the framework is chosen in document 3	Core	2 and 4
Design principles and patterns	NFR-MNT-01	SOLID, package-by-feature, Strategy for pricing and tax, domain events, Repository, DTO; an architecture test with ArchUnit	Core and Stretch	1-3

Topic	Needed for	What to be able to do	Level	Phase
Performance tuning	NFR-PERF	Reading query plans, index design, connection pool sizing, measuring p95	Stretch	3
Security hardening extras	FR-AUTH-06 and 07	Rate limiting, refresh-token rotation, secrets management	Stretch	3

If time runs short, cut Stretch topics and Should or Could requirements first. Do not cut the Core concurrency topic or its tests: they are what separates this project from a CRUD tutorial.

14. Open issues

ID	Question	Decide in
OI-01	Should BUMP be allowed directly from NEW (one tap in a busy kitchen)? Currently no.	Document 6, after UI wireframes
OI-02	Client UI technology: server-rendered pages or a small React app.	Document 3
OI-03	Database: PostgreSQL (default) or MySQL.	Document 3
OI-04	Confirm the default recall window (10 minutes) and colour thresholds (80 and 100 percent) against a realistic kitchen routine.	Document 6
OI-05	GST treatment: menu prices exclusive or inclusive of GST. The formula in BR-03 changes if inclusive.	Before pricing tests are written
OI-06	Does the DELIVERY order type need an address? Assumed no (label only).	Document 5
OI-07	Your deadline and company assessment details; these shape the roadmap, not the requirements.	Roadmap review

15. Glossary

Term	Meaning
QSR	Quick service restaurant
KOT	Kitchen order ticket, the paper slip the KDS replaces
KDS	Kitchen display system
Station	A kitchen work area such as Fry or Grill
Ticket	The part of an order handled by one station
Bump	Mark a ticket done
Recall	Undo a bump
86	Mark an item unavailable
RUSH	Priority flag that moves an order to the front of station queues
Token	Short number the customer watches on the status board
Business day	Operating day starting at the configured cutoff time
Idempotency	Repeating a request has the same effect as sending it once
Optimistic locking	Detecting concurrent edits with a version number instead of holding a lock
SSE	Server-sent events, a one-way stream from server to browser
JWT	JSON Web Token, a signed token that carries the user's identity and role
RBAC	Role-based access control
p95	The value under which 95 percent of measurements fall

Term	Meaning
GST	Goods and Services Tax

16. Revision history and next step

Version	Date	Author	Change
0.1	28 October 2026	Rathi	First draft

Next: document 3, System Architecture (HLD). It closes OI-02 and OI-03 and covers the module structure, the SSE flow, transaction boundaries, the deployment view and the main design decisions.